Assignment 1: Kinematics

ROSE5710: Fundamental Robotics and Systems | 6 November 2026

Modified D-H convention and leg chains

The transforms describe each site relative to the robot base. All translations below are in metres and all joint angles are in radians. The final row in each table is fixed.

```
A_i = Rx(alpha_(i-1)) Tx(a_(i-1)) Rz(theta_i) Tz(d_i)
theta_i = s_i q_i + theta0_i; T_B,site = T_B,0 A_1 ... A_n
```

Define r = sqrt(0.019² + 0.035²) = 0.03982461550348 m and beta = atan2(19, 35) = 0.49734271634682 rad. Code stores each row as [alpha, a, theta0, d]; these tables follow the question's [alpha, a, d, theta] order.

Left leg (5 joints + foot site)

`T_B,0 = Rz(pi/2 + beta); signs s = [1, -1, -1, -1, -1].`

i	alpha_(i-1) [rad]	a_(i-1) [m]	d_i [m]	theta_i [rad]
1	0	r	-0.0000741	q1 - beta
2	pi/2	0	-0.01606	-q2 + pi/2
3	pi/2	0	0.07415	-q3 + pi
4	0	0.07865	0	-q4
5	0	0.07865	-0.01975	-q5 - pi/2
6 fixed	pi/2	0.0005	-0.036225	0

Right leg (5 joints + foot site)

`T_B,0 = Rz(-pi/2 - beta); signs s = [1, -1, 1, -1, -1].`

i	alpha_(i-1) [rad]	a_(i-1) [m]	d_i [m]	theta_i [rad]
1	0	r	-0.0000741	q1 + pi + beta
2	pi/2	0	-0.01606	-q2 + pi/2
3	pi/2	0	-0.07415	q3 + pi
4	0	0.07865	0	-q4
5	0	0.07865	0.01735	-q5 - pi/2
6 fixed	pi/2	0.0005	-0.036225	0

Frame assignment: in the zero pose, the leg D-H joint axes z1...z5 are [+Z, +X, +Y, +Y, +Y]. Each x axis follows the common normal; virtual origins are selected at the yaw/roll and roll/pitch axis intersections. The final fixed row aligns the foot-site axes with the base axes at zero joint angles.

Geometry: the roll-axis height is 45.9409 - 46.015 = -0.0741 mm. Shifting origins along the parallel pitch axes combines the small CAD offsets into the specified final lateral distances: left 19.75 mm and right 17.35 mm. The two legs therefore require different final lateral offsets.

Head chain and inverse kinematics

Head and neck (4 joints + head site)

`T_B,0 has identity rotation and p = [0.001, 0.01915, 0.0900009] m; signs s = [1, 1, 1, 1].`

i	alpha_(i-1) [rad]	a_(i-1) [m]	d_i [m]	theta_i [rad]
1	pi/2	0	0	q1 + pi/2
2	0	0.066	0.01905	q2 - pi/2
3	-pi/2	0	0.03785	q3 + pi/2
4	-pi/2	0	-0.005	q4
5 fixed	pi/2	0	0.04245	-pi/2

In the zero pose the head D-H joint axes are [-Y, -Y, +Z, -X], matching the MuJoCo positive directions. The neck-to-head-pitch separation is 66 mm. The 92.021 mm base-to-neck distance is already included in T_B,0 and must not be added again. The effective vertical separation from the head-pitch axis to the roll axis is 37.85 mm; the net roll-axis offset is 5 mm.

Zero-pose cross-check

Site	Base-frame position [m]	Rotation
left foot	[-0.03456, 0.08940, -0.1935991]	I
right foot	[-0.03456, -0.09180, -0.1935991]	I
head	[0.00600, 0.00010, 0.2363009]	I

Numerical IK method

For a target pose (p_t, R_t) and its FK prediction (p(q), R(q)), the position residual is target minus prediction. The orientation residual is the axis-angle rotation vector, rather than an Euler-angle difference.

```
e_p = p_t - p(q); e_R = rotvec(R_t R(q)^T)
E(q) = ||10 e_p||² + ||e_R||²
g_i = [E(q + h e_i) - E(q - h e_i)] / (2h)
q_next = clip(q - alpha g, q_min, q_max)
```

Use h = 1e-6 rad, fixed alpha = 0.1, at most 5,000 iterations per target, and stop at E < 1e-8. Joint limits are read from MuJoCo. Each solve starts from the current joint angles, and only the selected chain is updated. The stopping rule bounds position error below 1e-5 m and orientation error below 1e-4 rad.

Supplied trajectories

Both feet move forward and upward by up to 20 mm at fixed orientation; the left uses $\sin^2$(pi phase) and the right uses $\cos^2$(pi phase). The head follows a neck-centred arc of 0.1 $\sin^2$(pi phase) rad, updating both position and orientation. The base stays fixed and every target is checked against MuJoCo site poses.

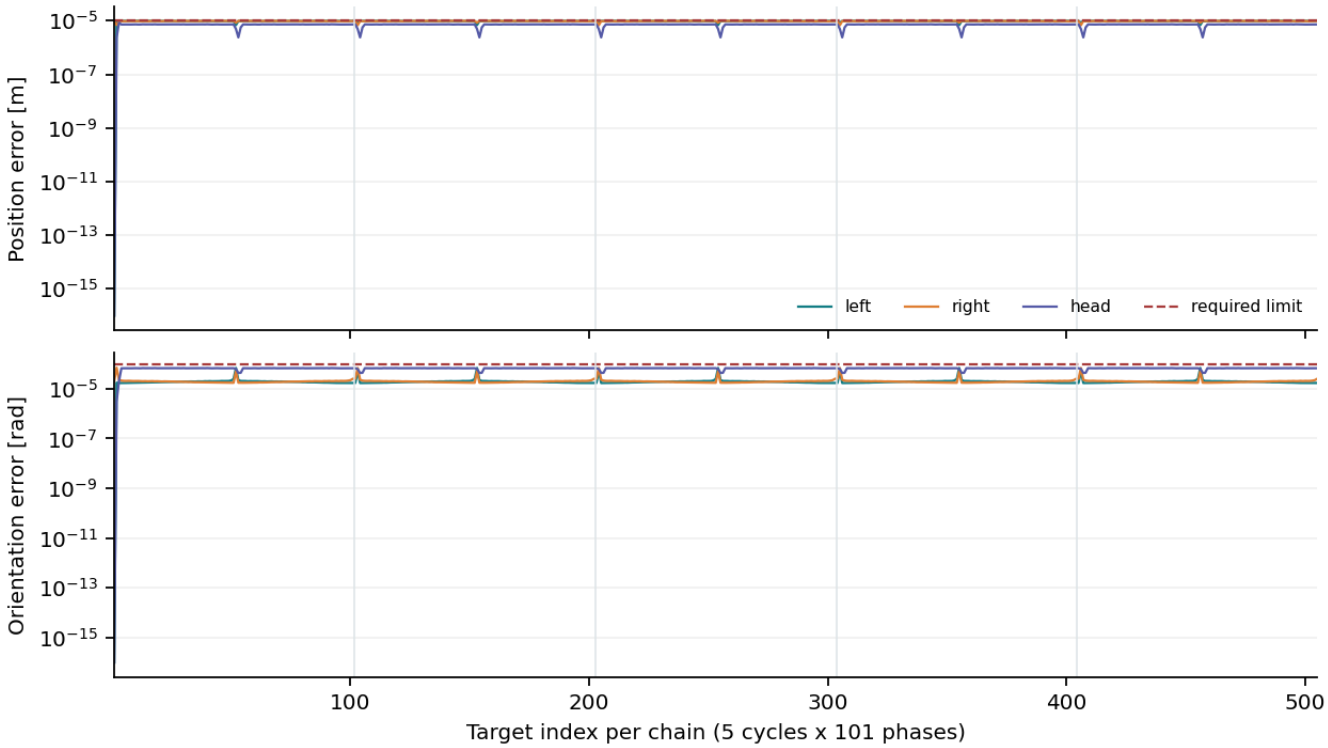
Measured verification results

Python 3.11.18 | NumPy 1.26.4 | SciPy 1.14.1 | MuJoCo 3.2.4

FK: 100 joint-limit-uniform configurations per chain with seed 0 (300 total). IK: all three supplied trajectories, 5 complete cycles and 101 phases per cycle (1,515 measured chain poses). Headless execution changes only rendering and delay.

Check	Chain	Max position [m]	Max rotation [rad]	Result
FK	left	8.450861e-15	2.285348e-14	PASS
FK	right	4.429243e-14	2.357637e-14	PASS
FK	head	6.572043e-15	1.032631e-13	PASS
IK	left	9.847942e-06	6.846493e-05	PASS
IK	right	9.844913e-06	7.038121e-05	PASS
IK	head	8.393151e-06	6.773237e-05	PASS

Required maxima: FK < 1e-6 m and < 1e-5 rad; IK < 1e-5 m and < 1e-4 rad. Every measured value passes. Six independent regression tests also cover base-frame invariance, pose-error weighting, reachable IK, joint limits, and preservation of other chains.



All 505 target indices per chain are shown. Dashed lines mark the required maxima; vertical lines separate cycles. Values below 1e-16 are floored only for the log plot. JSON files retain the original measured values.

Reproduce from the project root

```
python assignments/assignment_1_kinematics/forward_kinematics.py
python assignments/assignment_1_kinematics/inverse_kinematics.py --headless
python -m unittest assignments.assignment_1_kinematics.test_kinematics
```

Use the Python 3.11 environment and requirements.txt listed in README.md. The results directory contains the per-configuration FK data and all IK phase errors. Add --output-json to either script to regenerate its results; build_report.py rebuilds this PDF from those files.